

Proxy Embeddings: Why You Cannot Validate a Proxy by Correlating It With the Artifact

Abstract

A synthetic-data pipeline embeds text and then acts on the distances: drop the near-duplicates, keep the most different items, retrieve neighbours, screen for contamination. Every one of those decisions bets that proximity in the embedding stands in for proximity in the thing the text becomes — a program that runs, a query that returns rows, a picture that gets rendered. The usual way to check the bet is to decode a sample and report a correlation.

That check does not work, and we can say exactly why. On 553 generated Python programs decoded by execution on a fixed 140-input battery, text distance and behavioural distance correlate at +0.451 among the closest decile of pairs and +0.065 among the farthest — an apparently sharp result saying that embeddings track behaviour among near-duplicates and carry nothing beyond them. Running the identical analysis between two *encoders*, with no decoder anywhere, gives **+0.742 to +0.781** for the same near-minus-far quantity: **the decoder-free control shows the effect about twice as strongly as the decoder does.** The conditional-mean curves coincide. The shape is a property of comparing two distance matrices, not of the artifact. A permutation null does flatten (−0.025), so a real text-to-behaviour association exists; what does not exist is evidence for the mechanism the profile appears to show.

This is the check the field runs when it proposes an embedding proxy for something expensive, frequently with the observation that agreement is strongest among similar items. We give the control that should accompany it, and the positive alternative: **validate a proxy by running the decision, not by correlating the distances.** Scored that way, on decoders that are deterministic and total, three decisions come apart cleanly.

Deduplication is a corpus instrument, not an item gate. Duplicate-pair detection runs at AUC 0.779, but at the tightest radius tested **45.4% of the pairs a filter would reject are behaviourally far apart** (41.6% on a second decoder), so a filter defensible for contamination screening is not defensible per item, and no threshold fixes it.

Diversity selection has no stable sign. Greedy max-min loses to random at every budget on one decoder and beats it from $k = 20$ on another. It cannot be predicted from the embedding and has to be measured.

Both signs lose to the artifact-space oracle, which is cheaper than the proxy. The oracle reaches at $k = 30$ what the best text-space selector needs $k = 100$ to reach. Decoding a generated program in its own operating-system process costs 29.1 ms against 107.9 ms for a batched local embedding; tracing an `ipaddress` call costs 0.98 ms against 685 ms. For deterministic decoders the oracle is not the option a practitioner cannot afford but the one they did not think to run.

Text-level deduplication performed perfectly on this corpus removes 63% of it and leaves it **5.2× redundant in the space that matters**: 203 distinct programs, 39 distinct behaviours.

We ship all of it as `decodergap`, which reports the decoder-free control beside every correlation and scores each decision in the artifact's own space. Alongside this we report the same question where decoding is genuinely expensive — rendered images and audio — where text-embedding similarity explains 14.5% of the variance in rendered-image similarity, whether a prompt can be steered at all is a property of the embedder, and a third of the conditioning is lost at the seam where one model's language becomes another model's input.

1. Introduction

A generated corpus is almost never the thing anyone wants. It is a step towards the thing: the program that will run, the query that will return rows, the picture that will be rendered, the model that will be fine-tuned. Between the text and the artifact sits a **decoder** — a compiler, an interpreter, a database engine, a diffusion model — and the pipeline that produced the text does not consult it. It embeds the text and makes its decisions there.

Those decisions are the standard furniture of the field. Deduplicate at a similarity threshold. Keep the k most mutually distant items. Retrieve the nearest neighbours of a query. Check an evaluation set against a training set for leakage. Each is a claim that the embedding's geometry stands in for the artifact's, and each is made by pipelines that never test it, because testing it appears to require decoding everything, which appears to be expensive.

This paper tests it. The companion paper, *Recursive Axis Conditioning for Diverse Synthetic Data Generation*, builds a generation loop on an explicit assumption — an embedding oracle E mapping an item to a point in $\mathbb{R}^D$, with no inverse — and every axis it scores, every candidate it selects and every number it reports is a statistic of E . This paper is about the word *oracle*. In two of that paper's five domains the artifact is not text at all, and even where it is, the embedder is blind to metre, to compositional template, and to where an exam item's answer key sits among its options. The oracle is always a proxy. What we add here is that the question generalizes far past diversity, and that in the domains where the decoder is deterministic it has a clean and repeatable answer.

What we found first was a boundary that looked clean: embeddings appearing to be **near-field instruments**, separating near-duplicates from everything else and saying little about which of two already-distinct items is more distinct. That would have explained a great deal — why deduplication reliably works, why diversity selection reliably disappoints. It did not survive its own control. The near-field shape appears just as strongly between two encoders that have never seen the artifact, so it is a property of comparing distance matrices rather than a fact about decoders, and we report it here as the confound it is rather than the mechanism we wanted.

What survives is more useful and less comfortable. Correlating an embedding against an artifact measurement is not a valid way to qualify a proxy, and the alternative is to run the decision itself and score it where the artifact lives. Done that way the decisions separate: deduplication survives at corpus scale and fails per item, diversity selection changes sign between decoders, and the artifact-space oracle beats both while costing less than the embeddings it replaces.

Choosing deterministic decoders also removes a bound that no amount of care removes otherwise. The image endpoint used by the companion paper exposes no seed, so no render can be repeated and render noise cannot be separated from what is being measured; every artifact-space number there is a statement about an unlogged seed distribution. A Python program on a fixed battery, a query against a fixed database, a pattern against a fixed corpus of strings: each repeats exactly, so artifact distance is a property of the text alone.

The paper is in two halves. §4 runs the control that invalidates correlational proxy validation and then measures the three decisions directly on deterministic decoders, §9 measures what decoding

costs against what embedding costs, and §10 describes the probe we ship. The second half is the expensive-decoder case, where a render or a two-minute audio generation sits between the text and the artifact and the oracle is genuinely out of reach. **What the proxy sees** (§5): text-embedding similarity between image instructions predicts rendered-image similarity at Pearson r between 0.167 and 0.421, so most of the visual variance is unexplained by anything a text-side method optimizes; in audio, whether a prompt can be steered before rendering is a property of the embedder, with prompt-to-track alignment of 0.68 under MuQ-MuLan against 0.18 under CLAP-music; literal and latent diversity of one corpus move in opposite directions as it grows; a whole class of visual repetition — tiled grids, shared palettes, recoloured compositions — is invisible to embedding metrics and visible to a 16×16 luminance map; and a generated exam bank whose keys sit 90% in position A scores identically on every diversity measure to a balanced one. Against that, the *ranking* of methods does survive the choice of viewpoint: under four independent representations, three of which the method never optimizes, RAC ranks first under every one. **Steering through the proxy** (§6): rendering a bounded sample, embedding it in the artifact's own space and mapping its crowded directions back into the text space gives the best image arm at matched n and, through an audio embedder, raises mean-centered CLAP Vendi on rendered tracks from 11.5 to 17.43; and when a pipeline pays for extra renders, choosing among them on the channel the objective cannot see buys $1.24\times$ on that channel at zero cost in CLIP. **Conditioning across the seam** (§7): by intervention, nine of eleven image axes are realized in the written instruction and four of eleven in the render, a mean loss of 35% of the effect at the boundary where one model's language becomes another model's input. **Scoring in the artifact's space** (§8): the obvious way to make axis scoring empirical — replace level descriptions with the centroid of what each level produced — is a marginal mean where a partial effect is needed, and it is the companion's manipulation-based repair that makes artifact-space scoring usable at all.

The contributions, in the order we would defend them:

- **A confound in proxy validation** (§4.3): the near-field agreement profile that appears to qualify an embedding as a proxy is reproduced, more strongly, by two encoders with no decoder involved. With the control that should accompany any such correlation, and the permutation null that shows what a real association looks like.
- **A decision-by-decision verdict** (§4.5–§4.7), each a direct measurement that the confound does not touch: deduplication supported at corpus scale and refused per item at a 45.4% false-positive rate, diversity selection sign-unstable between decoders, and $5.2\times$ behavioural redundancy surviving perfect text-level deduplication.
- **The cost inversion** (§9): for deterministic decoders the artifact-space oracle is cheaper than embedding the corpus, by $3.7\times$ in the least favourable measurement we can make and by three orders of magnitude in the most.
- **decodergap** (§10), the probe that reports all of the above for a corpus, an embedder and a decode function, and that declines to recommend a threshold.
- **A measured proxy gap in three modalities** (§5.2, §5.3), with the pooled and within-arm shared variance stated separately and the embedder named in every audio number.
- **Cross-modal steering as an extension of the loop** (§6.1, §6.2): the same ledger that repels what the model keeps *saying* also repels what it keeps *showing*, through a least-squares bridge from the artifact's embedding into the text embedding the loop already optimizes.
- **The seam result** (§7): conditioning strength degrades where the generator changes hands, established by manipulation rather than correlation.
- **Two failures no embedding can see** (§5.6, §5.7), with the repair for each that needs no embedding.
- **Robustness of the ranking to the viewpoint** (§5.4): four independent representations agree at a mean Kendall τ of +0.83.

2. The oracle is a proxy

We assume throughout:

Assumption (embedding oracle, no inverse). We have access to E : $\text{Text} \rightarrow \mathbb{R}^D$, computable on demand. We have **no** E^{-1} : $\mathbb{R}^D \rightarrow \text{Text}$.

This is the structural fact that shapes the entire design. Given a corpus X_n we can compute, exactly and cheaply, the point in $\mathbb{R}^D$ that would most improve any of our diversity measures — the direction of least occupied spectral energy, the centre of the largest empty ball, the point maximizing marginal coverage. Knowing that point is worth nothing on its own, because no procedure turns a target embedding back into a poem.

Consequently the system cannot *solve* for its next item. It can only **propose** (sample from $p(\cdot | x)$ for some prompt x we can write), **measure** (embed the proposals and score them), and **select** (keep one). All of the control is in the choice of x , and that choice is expressible only in language. This is why the latent variables here are language-valued — named axes with named levels — rather than continuous codes: they are the only handles that reach the generator. It is also why the axis scoring of the companion paper's §5 exists. **The axis scoring is a surrogate for the missing inverse:** unable to decode the direction we want to travel, we score the language-valued conditions we *can* write by how nearly their induced output distributions point that way.

Three consequences follow for measurement, and they order the evidence in this paper.

Two facts about circularity are worth stating before the results. Our packing selection rule maximizes a weighted sum of embedding-space orthogonality and embedding-space min-gap, and we then report embedding-space diversity metrics; centered Vendi is a monotone function of how flat the Gram spectrum is, which is close to what the orthogonality term climbs, and median nearest-neighbour distance *is* the min-gap term. Those wins should be read as confirmation that the optimizer works, not as independent evidence. The independent evidence is the literal measures, which the method never observes, and the out-of-objective channels of §5.1, which live downstream of a rendering step or in a feature space nothing in the loop touches. Sorted by how much weight they can bear: embedding-space measures are partly circular; literal-space measures are independent; rendered-artifact and out-of-objective measures are the most independent, and they carry the argument.

The proxy adds a fourth rung below all three. When the artifact is rendered by a model the embedder never sees, even the most independent text-side measure is a measure of the *instruction*, and whether the instruction's diversity reaches the picture is an empirical question — with, it turns out, a discouraging answer (§5.2). The companion paper states five practices that follow from naming the objective; this paper is where the last three are tested.

1. **Say which objective you mean.** They are different problems with different optimal policies, and "diverse" does not distinguish them.
2. **Count exact duplicates before computing anything.** A 74% duplicate rate makes every distance statistic a statistic about duplication.
3. **Publish the pairwise-similarity distribution** your kernel operates on. Mean pairwise cosine was 0.883 in one of our corpora and 0.444 in another under the same embedder; an ϵ or a Vendi score means nothing without it.
4. **Measure at the level of the artifact you ship.** Text-embedding similarity explains 14.5% of the variance in whether rendered images look alike, pooled across seven rendered arms, and 8.2% within-arm (§5.2).
5. **Audit in a channel the objective never optimizes** (§5.1, §5.7; the companion paper's §8.3). It is the only measurement in this paper that cannot be a restatement of the selection rule, and it is where both the strongest positive result and the largest undetected defect were found.

3. Setup

Generator and embedders. openai/gpt-5.6-luna via OpenRouter for generation, judging, attractor mining and axis elicitation. Text embeddings: nomic-embed-text (768-d, unit-normalized) served locally by Ollama, so embedding is free and the loop is never rate-limited by its own measuring instrument. Images: gpt-image-1-mini, embedded with CLIP ViT-B/32. Audio: Lyria 3 Pro instrumentals, embedded with CLAP and MERT. Every corpus is append-only JSONL with embeddings checkpointed alongside, resumable after a kill; every number carries a provenance tag naming the procedure that produced it.

Domains. *DALL-E instructions for post-modern artworks*, where diversity is the product and the

same corpus can be measured in three spaces — words, text-embedding, and the rendered picture. *Psychometric test questions*, multiple-choice items assessing general reasoning, where two items with the same construct are a validity problem rather than an aesthetic one. *Instruction corpora*, for the head-to-head against released artifacts. *Poetry*, a pilot where craft rather than content carries the variation. *Instrumental-music prompts*, the longest chain in the paper: latent axes → text prompt → two-minute instrumental → audio embedding.

Methods compared. Five published approaches, implemented faithfully rather than as strawmen, each getting the same generator, embedder and budget accounting as ours.

arm	what it is
naive	plain repeated prompting — the honest floor
high_temp	temperature 1.6 — the trivial diversity lever
self_instruct	Self-Instruct / Alpaca : few-shot exemplars sampled from the pool, plus ROUGE-L ≤ 0.7 rejection against it
evol_instruct	Evol-Instruct (WizardLM) : sample an existing item, apply a random evolution operator (deepen, concretize, add constraint, harder reasoning, mutate form)
persona	Persona-Hub / AttrPrompt : a flat catalogue of personas $\times$ attributes, sampled uniformly
rac	Recursive Axis Conditioning (ours)
rac+vision	RAC, plus steering on the <i>rendered image</i> (§6.1)

What we measure, and what each measure misses. We report several because they disagree, and a corpus that one of them calls healthy another calls collapsed.

- **Exact-duplicate rate** — byte-identical repetition. Unambiguous, cheap, and the first thing to compute; blind to paraphrase.
- **Distinct- n , 4-gram self-repetition, n -gram Vendi** — surface statistics over token sequences. They catch templating that the duplicate rate misses and read meaning not at all. **These are independent of our objective**, since the method sees only embeddings and has no access to token counts or string identity.
- **Mean-centered embedding Vendi** — the exponential of the von Neumann entropy of the corpus Gram matrix, an effective number of distinct items. It summarizes the whole spectrum, which makes it insensitive to local structure. Centering matters as much as the statistic: the shared mean direction of text embeddings compresses the uncentered score by roughly $2.7\times$, so an uncentered Vendi reports the cone as much as the content.
- **Median nearest-neighbour distance** — how much room the typical item has. It goes to exactly zero when the median item has a perfect twin.
- **Worst-case nearest-neighbour distance (the packing radius)** — the only measure here under which a single collision is a defect regardless of everything else.
- **Coverage, density, precision and recall against a reference** — computed with reference-side k -NN radii so nothing is tunable per corpus. These are the only measures that know what the space is supposed to look like, and the only ones that let corpora of different scales be compared. Covered fraction at a *fixed* radius does not survive that comparison: it correlates -0.991 with within-corpus spacing, so it ranks corpora by how tightly they cluster rather than by how much they reach.
- **A judged threshold**, where an application defines the failure (the companion paper's §7.4).
- **Measures on the rendered artifact, and measures in a channel the loop never optimizes** (§5.1), which are the ones that cannot be circular. **Scale and cost.** The text corpora comprise 43,171 real generations for roughly \$8.50 of OpenRouter spend, plus 1,796 rendered images across sixteen image arms and up to three quality tiers, a further 285 renders for the two render-side probes of §6.3 and §7, 100 rendered Lyria instrumentals, and 236 adjudicated exam-item pairs. Both naive arms reach $n = 10,000$; the reimplemented baselines reach 2,500 each. Every comparison is reported at a matched n that all compared arms actually reached. Total API spend across the project is roughly \$40.

4. A proxy cannot be validated by correlating it with the artifact

Everything in this paper so far is a statement about diversity. The measurement underneath it is not. A pipeline that embeds text and acts on the distances is making a bet that has nothing to do with diversity in particular: that proximity in the embedding stands in for proximity in the artifact. Deduplication makes it, contamination screening makes it, nearest-neighbour retrieval makes it, coreset selection and active learning make it.

The natural way to check the bet is to decode a sample, compute a distance in the artifact's space, and correlate. This section shows that this check does not work — not that it is noisy, but that it returns the same answer when the artifact is replaced by something with no connection to it — and then measures the three decisions directly instead, which is what survives.

4.1 Decoders that leave nothing to interpretation

An image is rendered by a model, which is why every artifact-space number in the companion paper is a statement about an unlogged seed distribution: the endpoint exposes no seed, so a render cannot be repeated and render noise cannot be separated from the quantity being measured. That bound is structural.

It is removed by choosing decoders that are *deterministic and total*. A Python program run on a fixed battery of inputs, a SQL query run against a fixed database, a regular expression matched against a fixed corpus of strings: each maps text to an artifact with no sampling anywhere, so artifact distance is a property of the text alone and repeats exactly. These are also among the largest synthetic-data industries there are.

domain	text	artifact	artifact distance	coverage
code	a Python function $f(xs)$	its results on a fixed 140-input battery	fraction of the battery on which two programs disagree	distinct (input, result) cells
SQL	a SELECT against a fixed schema	the rows it returns	Jaccard distance between result-row sets	distinct rows retrieved
regex	a pattern	the subset of a 210-string probe corpus it matches	Jaccard distance between matched sets	distinct strings matched
library inputs	a string handed to <code>ipaddress</code>	the arcs of the library it executes	Jaccard distance between arc sets	distinct arcs reached

None of these distances is computed with reference to an embedding, so each can adjudicate an embedding rather than agree with it. The first is the corpus used below: 553 generated Python programs, of which 203 are distinct sources, which between them exhibit **39 distinct behaviours**.

4.2 The profile that looks like a finding

Correlating text distance against behavioural distance *within* each decile of text distance, rather than pooling, produces a table that appears to say something sharp:

decile of text distance	width	pairs	corr. with behavioural distance	mean behavioural distance
1 (closest)	0.140	2,051	+0.451	0.549
2	0.032	2,050	+0.112	0.827
3	0.022	2,050	+0.010	0.877
4–9	0.014–0.018	12,301	−0.040 ... +0.062	0.878–0.909
10 (farthest)	0.104	2,051	+0.065	0.916
pooled		20,503	+0.364	0.853

Read on its own this says: the embedding tracks behaviour among near-duplicates and stops carrying information beyond them. The conditional mean of behavioural distance given text distance says the same thing without any correlation at all, rising **+0.474 across the near half of the distance range**

and **+0.036 across the far half**. Every diversity selector in the literature draws its picks from the top decile.

We believed this, and it is wrong.

4.3 The control: two encoders, no decoder

The profile compares one distance matrix to another and reports that they agree more among close pairs. Nothing in that description mentions a decoder. If any two distance matrices over the same points agree preferentially in the near field, the shape is a property of the comparison and the decoder is doing no work at all.

Running the identical profile between two *encoders*, with no decoder anywhere in the computation:

comparison	near-field corr.	far-field corr.	near – far
nomic vs behaviour (<i>the claim</i>)	+0.451	+0.065	+0.386
char TF-IDF vs behaviour	+0.474	–0.002	+0.476
word TF-IDF vs behaviour	+0.513	–0.053	+0.567
nomic vs char TF-IDF (control)	+0.742	–0.003	+0.745
nomic vs word TF-IDF (control)	+0.750	+0.007	+0.742
char TF-IDF vs word TF-IDF (control)	+0.898	+0.117	+0.781
nomic vs <i>permuted</i> behaviour (<i>null</i>)	–0.002	+0.023	–0.025

The decoder-free controls show the shape about twice as strongly as the decoder does. The conditional-mean curve is the same story: the control rises +0.493 over the near half and +0.030 over the far half, against the decoder's +0.474 and +0.036 — the two curves are the same curve. Two representations that have never seen a program run produce a cleaner "near-field law" than the comparison against what the programs actually do.

A third of the effect is bookkeeping. Equal-count deciles have very unequal widths — the first spans 0.140 of the distance range and the eighth spans 0.014 — and a within-bin correlation is attenuated by the bin's own spread. On equal-width bins the profile is +0.288, +0.056, +0.114, +0.191, +0.061, –0.004, +0.014, +0.030, +0.070, –0.080: still more signal at the low end than the high end, and nothing like the monotone decline the decile table displays.

One control does pass, and it is the one that matters for what remains. Permuting the artifacts across items — breaking the text-to-behaviour correspondence while leaving both marginal distance distributions exactly intact — flattens everything to –0.025 near-minus-far and –0.003 pooled, against +0.364 unpermuted. **There is a real association between what a program looks like and what it does.** What there is no evidence for is the shape we read into it.

4.4 The consequence: correlational proxy validation is confounded

This generalizes past our own mistake, because the check we ran is the check the field runs. A paper that proposes an embedding-based proxy for something expensive — human preference, rendered quality, downstream accuracy, behavioural diversity — validates it by decoding a sample and reporting a correlation, and frequently reports that the correlation is strongest among similar items. That last observation is not evidence. It appears between two arbitrary encoders that have never seen the outcome.

The prescription is cheap and almost nobody follows it:

Report a decoder-free control beside any proxy-validation correlation. Correlate your embedding against a second, unrelated representation of the same corpus. Whatever agreement survives *above* that control is the part your artifact measurement contributed. On the corpus here, that residue is negative: the decoder profile is weaker than the encoder-to-encoder baseline.

And the positive form: **validate a proxy by running the decision, not by correlating the distances.** A correlation is a summary of a geometry; a decision is a thing that either works or does not, is scored in the artifact's space, and has no confound of this kind. The rest of this section does that for the three decisions a synthetic-data pipeline actually makes.

4.5 Decision 1 — deduplication is a corpus instrument, not an item gate

Deduplication reads the closest pairs, which is where whatever signal exists is concentrated, and at corpus scale it works: duplicate-pair detection on the code corpus runs at **AUC 0.779** against the ground-truth label *identical on the whole battery*.

Per item it does not. Among the pairs a near-duplicate filter would reject, the fraction that are behaviourally *far apart* (distance > 0.5):

filter radius	pairs rejected	of those, still behaviourally far	mean behavioural distance
5th pctl	1,026	45.4%	0.419
10th	2,051	62.2%	0.549
20th	4,101	76.8%	0.688
30th	6,151	83.0%	0.751
40th	8,201	86.3%	0.784
<i>pool-wide</i>			<i>0.853</i>

At the tightest radius, nearly half of what the filter discards behaves unlike the item it was discarded for resembling; the best radius available reaches F1 0.449 (precision 0.452, recall 0.446). A second decoder — 202 strings scored by the `ipaddress` arcs they execute — gives 41.6% at the same quantile against a pool-wide mean of 0.672.

This is a direct measurement of the decision, so the §4.3 control does not touch it. Threshold tuning does not fix it either, because the error is in the strength of the evidence rather than the location of the boundary. **A text-space duplicate filter is defensible as a corpus-level instrument — a duplicate rate, a contamination screen — and not as a per-item accept/reject.**

4.6 Decision 2 — diversity selection has no stable sign, and both signs lose to the oracle

Farthest-point traversal draws its picks from the top decile. What that costs turns out to depend on the decoder, and not in a way anything visible in the text predicts. On the `ipaddress` decoder, greedy max-min is worse than random at every budget in every representation, by 54 to 82 arcs. On the code corpus it is worse than random at $k = 10$ and better from $k = 20$ on:

k	random	max-min	best near-duplicate filter	artifact-space oracle
10	815	743	849	1,336
20	1,119	1,296	1,267	2,047
30	1,355	1,509	1,517	2,365
50	1,549	1,864	1,809	2,414
100	1,891	2,358	2,113	2,414

Behaviour cells covered, mean of 20 seeds; the oracle is greedy marginal coverage computed in the artifact's own space.

Two things follow. **Whether text-space diversity selection helps is a property of the decoder and is not predictable from the embedding** — two decoders, the same selector, opposite signs. And **the gap that matters is against the oracle, not against random**: the best text-space selector at $k = 10$ buys 849 cells where the oracle buys 1,336, the oracle reaches at $k = 30$ what text-space selection needs $k = 100$ to reach, and it saturates the pool at $k = 50$ where no text-space arm saturates at all. Text-space selection leaves between a third and a half of the reachable behaviour unbought at small budgets whether or not it beats random.

The oracle is normally waved away as the thing one cannot afford. §9 measures what it costs.

4.7 Decision 3 — the redundancy that no text-space instrument reaches

The corpus itself makes the case for measuring the artifact more compactly than any correlation does. Naïve repeated prompting produced 553 programs. Removing byte-identical sources — the

deduplication step every pipeline performs — leaves **203**. Those 203 distinct programs exhibit **39 distinct behaviours**.

So text-level deduplication, performed perfectly, removes 63% of the corpus and leaves it **5.2× redundant in the space that matters**. The residue is not subtle paraphrase. One pair of programs in it implements the longest-increasing- subsequence length by patience sorting with a binary search, and the other by the quadratic dynamic program; they share almost no tokens, and they agree on all 140 battery inputs. No embedding-space threshold reaches that pair without also discarding most of the corpus, and running both programs takes microseconds.

5. What the proxy sees

5.1 Audit in a channel the proxy never sees

The strongest evidence available for any embedding-optimized method is a channel that is *causally downstream of the artifact and upstream of nothing the loop reads*. Each artifact domain has one, and the companion paper's central comparison (its §7.5) is made there: twenty-two prosodic and structural features for poems, where nomic-embed-text is nearly blind to metre and rhyme; thirteen item-architecture features for exam items; nine pixel statistics — luminance, contrast, tonal range, saturation, colourfulness, hue entropy, edge density, edge anisotropy, and where the visual weight sits — for renders. At matched generator budget RAC separates from the strongest comparison on all three: 2.25× the prosodic spread of naive prompting on 200 of 200 random draws, 97.6% distinct item architectures against 43.9% after deduplicating both banks, and 1.53× the optical spread with every one of eleven RAC image corpora above every one of five published baselines ($U = 55$ of 55, $p = 0.0002$).

What that comparison teaches about instruments is the part that belongs here. **Use a distance statistic, not a spectral one, on a designed feature channel.** Vendi is the exponentiated entropy of a covariance spectrum and scores the effective *dimensionality* of variation rather than its extent; on a 768-dimensional embedding no single direction dominates and the distinction rarely bites, but on a nine-dimensional feature vector it dominates completely, and a corpus that spreads far along two or three optical directions scores *below* one that wobbles slightly along all nine. On mean pairwise distance, optical spread correlates with CLIP diversity at $r = +0.76$ ($p = 0.003$) across thirteen corpora; under the spectral estimator the same corpora correlate at -0.60 . The proxy and the artifact channel agree or disagree depending on which statistic is read off the channel, before any question about the embedding arises. Degradation does not explain the separation either: blurring k of sixty baseline renders lowers CLIP Vendi monotonically, 43.81 to 41.72 as k runs 0 to 30, so an embedding-space objective is penalized rather than rewarded for admitting degraded images.

The rest of this section is what such channels reveal that the proxy does not.

5.2 Text diversity is nearly blind to image diversity

We rendered the first 199 DALL·E instructions from the naive corpus and embedded them with CLIP. Pairwise cosine similarity in text-embedding space correlates with pairwise cosine similarity in image-embedding space at **Pearson r between 0.167 and 0.421** across the seven rendered arms, and at 0.285 within-arm centred. The gap varies by a factor of 2.5 depending on which corpus is measured: lowest on self_instruct (0.167) and on the naive arm (0.170), highest on the arm this paper's own method produced (0.421). Pooled, the shared variance is 14.5%; within-arm it is 8.2%. Knowing that two instructions are semantically far apart therefore tells you rather little about whether the two pictures look different, and every text-side diversity method in this literature — ours included — is optimizing a proxy that leaves most of what the reader receives unexplained. That the correlation is strongest on our own corpus is worth stating plainly: the proxy is least misleading exactly where the instructions were built to differ structurally rather than only lexically.

The qualitative version is more damning than the correlation. Independently generated instructions, from a corpus with a 0.000 exact-duplicate rate and healthy lexical diversity, render to near-interchangeable pictures: the same magenta-and-cyan palette, a classical marble bust, neon signage, halftone collage, a receding grid. A vision judge shown samples rates the set's distinctness 6–7 out of 10 and names the attractors precisely — *"muted beige, cream, brown, ochre and black foundations*

accented by saturated cyan/teal, turquoise, pink”, "appropriation of canonical or religious imagery, especially Mona Lisa-like female portraits", "frontal, museum-like presentation with centered, symmetrical compositions". This is a second mode collapse, downstream of ours, contributed by the image model and by the fact that much of what varies in the text lands in the same visual place.

Sixteen renders per policy, same generator, same budget, same image model

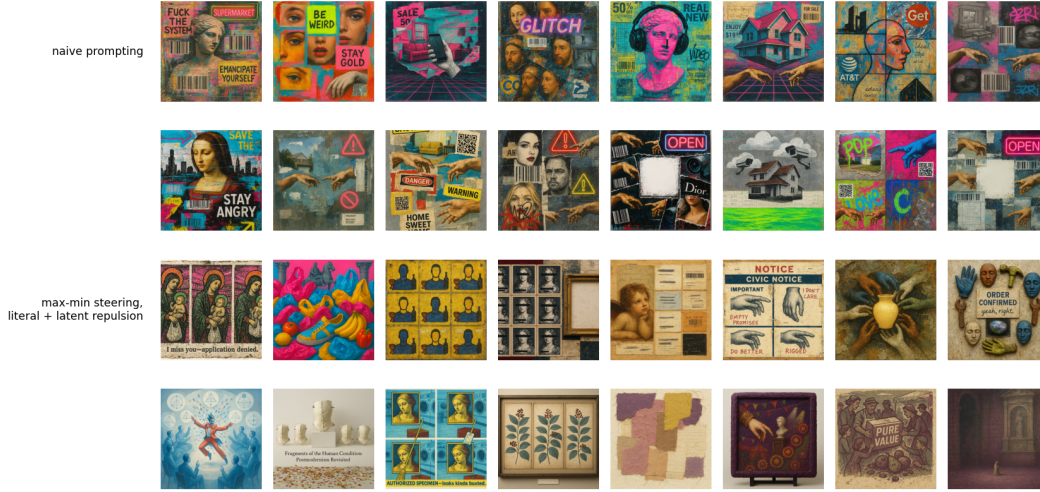


Figure 1. Sixteen renders per policy, same generator, same budget, same image model. Rows 1–2, naive prompting: a corpus with a 0.000 exact-duplicate rate and healthy lexical diversity that still returns one visual mode — magenta and cyan collage, barcodes and QR codes, Renaissance portraits and classical busts, Michelangelo hands, warning triangles and neon OPEN signs recurring across nearly every panel. Rows 3–4, max-min steering with literal and latent repulsion on both the text and vision sides: medium, palette, register and composition all move, across a medieval triptych, a botanical cabinet, a photographed sculpture installation, a torn-paper abstract and a civic notice.

5.3 Audio: whether a prompt can be steered is a property of the embedder

The more portable finding is that whether a prompt can be steered before rendering is a property of the *embedder* rather than of music. Measuring the Spearman correlation between prompt-pair similarity through the text tower and track-pair similarity through the audio tower:

embedder	alignment (naive / conditioned)	note
MuQ-MuLan	0.68 / 0.47	a usable steering gradient
CLAP-fused	0.50 / 0.21	weak; ~ 0.05 on the steered corpus
CLAP-music	0.18 / 0.06	worst; median within-corpus NN distance 0.004 — it hears one track

Cross-embedder agreement on pairwise track similarity spans 0.22–0.84, per-arm diversity verdicts flip between embedders, and each embedder nominates a different most-redundant pair. Two prescriptions follow: steer music with MuQ-MuLan rather than CLAP, and never publish an audio-diversity number without naming its embedder. Structural contracts have a length ceiling as well — compliance with a requested section sequence is exact at prompt lengths near 560 characters, 0.11 at a mean of 658, and zero by $\sim 1,600$ — so a structural contract for music must be short or it is noise.

5.4 The ranking does not depend on the viewpoint

A max-min score is a distance in a chosen embedding, so a natural worry is that it names a property of the embedder rather than of the corpus. Six exam-item corpora — this method's bank against naive sampling, persona prompting, self-instruct, evol-instruct and high-temperature sampling, 200 items each — were scored under four independent representations: CLIP text, TF-IDF with no learned

semantics at all, the twenty-two-feature prosodic vector, and a 150-dimensional function-word profile of the kind used for authorship attribution. Each score is the fifth-percentile nearest-neighbour distance divided by the mean pairwise distance of the same cloud, which is invariant to the global rescaling an embedding choice is free to apply, and each representation is admitted only if its between-corpus spread exceeds its own within-corpus sampling noise — all four clear that bar by factors of 9.7 to 24.7.

The four views agree, at a mean Kendall tau of +0.83 across their six pairings, and **RAC ranks first under every one of them**, including the three it never optimizes. It also holds the best worst-case score, so it wins under a distributionally-robust reading of diversity as well as under any single view. The agreement is not an artifact of the degenerate baselines: restricted to the three corpora whose fifth-percentile distance is non-zero in every representation, mean tau is +0.67 and RAC is still first under all four. Naive sampling and high-temperature sampling score exactly zero on prosody and on stylometry — at least one item in twenty has an exact neighbour in those channels.

5.5 Literal and latent diversity move in opposite directions

Literal and latent diversity move in opposite directions in one corpus, which is why both are reported throughout. Measuring the DALL·E naive corpus as it grows:

n	distinct-2	4-gram self-repetition	n -gram Vendi	centered embedding Vendi
5	0.870	0.031	4.9	3.79
40	0.492	0.211	30.2	24.13
300	0.252	0.384	142.5	55.70
1,000	0.145	0.520	246.6	63.05
1,750	0.112	0.577	283.3	65.30

Literal diversity collapses monotonically — by $n = 1,750$ more than half of each new instruction's 4-grams have already appeared — while latent diversity rises and then saturates. Reporting either alone supports an opposite conclusion about the same corpus.

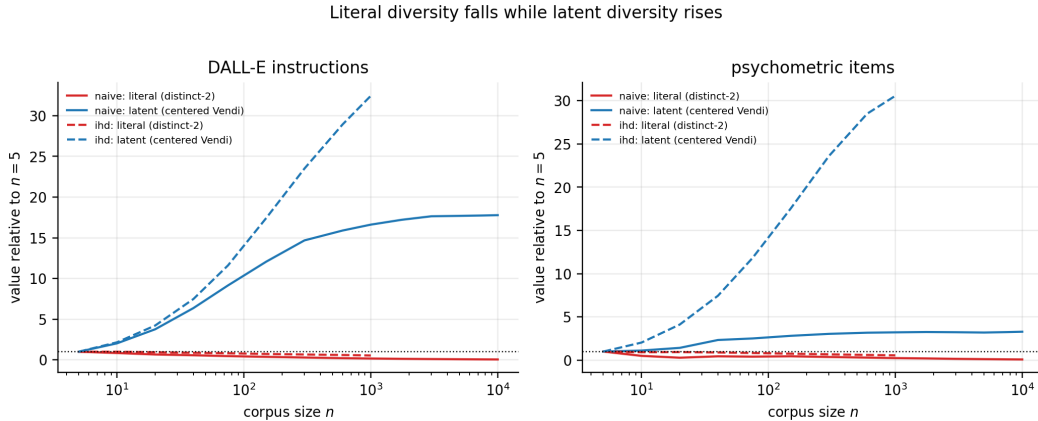


Figure 2. The decoupling, normalized to each series' value at $n = 5$. Literal diversity falls monotonically while latent diversity rises: two measurements of two different quantities that one word, "diversity", has been covering for.

5.6 A class of repetition the embedding misses

One comparison in the artifact channel runs against the method, and it belongs here. Embedding metrics miss an entire class of visual repetition — tiled grids of one cell, a shared palette, the same composition recoloured — so we audit it with deliberately dumb non-semantic signatures: a 16×16 luminance layout map, an autocorrelation tiling score, a hue histogram. Scored at matched $n = 60$ and matched render tier, **the published baselines are better than our arms on both structural measures**: layout twins above 0.5 cosine average 0.223 across the five baselines against 0.473

across our eleven, and literal tilings 0.117 against 0.406, with no baseline worse than our best arm on layout. A corpus told to differ along seven or eleven named dimensions apparently converges on a compositional template that plain repeated prompting does not. Adding literal-space structural bans to the prompt — grids banned when recent renders tile, dominant hue pairs named and banned, layout-change demands when layouts collide, plus a learned text→bad-structure bridge — halves palette twins (0.78 → 0.35 in the coverage arm) and improves the coverage objective 21% (0.206 → 0.247), but does not close it. Optical spread and structural repetition are both measured on pixels and point in opposite directions: our corpora occupy a wider region of that space while repeating their compositional scaffolding more often within it.

5.7 A defect no diversity measure can see

The audits of §7 ask whether the axes reach the artifact. This one asks whether the resulting bank is *usable*. For a test bank, key position is a hard requirement: correct answers must be distributed across option positions, because an examinee who notices an imbalance can exploit it without reading anything. Asking a fixed solver each item and recording which position it chose:

bank	A	B	C	D	χ^2 vs uniform (3 df)
RAC	0.900	0.050	0.050	0.000	90.4
naive	0.225	0.550	0.225	0.000	24.6
RAC, after balancing	0.200	0.175	0.300	0.325	2.6

Ninety percent of the RAC bank's keys sit in position A, against a critical value of 7.81 at $p = .05$. An examinee who answers A to everything scores 90% without reading a stem, and neither bank ever places a key in D. This is a property of the solver only if the solver is not reading the items, so we separated the two by permutation: ask each item twice, once as written and once with its options reordered, and record which option *text* is chosen. The answer follows the content in 95% of RAC items and 100% of naive items, and stays on the same letter in only 17.5% and 30%. The solver reads; the imbalance is in the bank.

Why the method cannot see this. Key position is not a semantic property, so it is invisible to every measure in this paper — embedding diversity, n -gram diversity, Vendi, coverage and the enemy-item radius are all exactly as favourable on a bank whose key is always A as on a balanced one. It is invisible to the axis set for the same reason: all seven axes condition on item content, and even *distractor logic* governs what the distractors are like rather than where the key sits among them. And it is structural rather than accidental — a generator asked for a stem and four options writes the answer it has in mind first and builds distractors around it, so A is where the key lands by default.

The repair needs no answer key. Permuting each item's options uniformly at random moves the key with its own text from whatever position it held, so the bank becomes uniform by construction; options whose text refers to a position ("none of the above", "both A and B") are detected and left as written, which affects 37 of 1,999 items. Applied to the RAC bank this takes χ^2 from 90.4 to 2.6 — indistinguishable from uniform — while content-tracking is unchanged at 95%.

The general form is the sharpest limitation in this paper. **A diversity objective defined over an embedding is blind to any property of the artifact the embedding does not encode, including properties that decide whether the artifact can be used at all.** Difficulty is the same story in the same domain: it is the parameter an item bank exists to span, it is latent rather than semantic, and nothing in the objective can see it. Measuring diversity well is not the same as measuring fitness for purpose, and the gap between them is not visible from inside the objective.

6. Steering through the proxy

6.1 Images: closing the loop where the product lives

The vision-steered arm closes the loop where the product lives. It renders a bounded sample of accepted instructions, embeds them with CLIP, and feeds two things back into the text-side loop: a least-squares map from the crowded *image* directions into instruction-embedding space, so the text-side orthogonality term can push away from visual redundancy it cannot itself perceive; and

mined *visual* attractors from the vision judge, appended to the same ledger as the textual ones. It is the paper's mechanism applied one level down — the ledger already repels against what the model keeps saying, and now also against what it keeps showing — and it is the best arm in the companion paper's competitive comparison (its §7.2).

The margin is measurable in the space the loop optimizes and visible on the page. At matched $n = 200$ in the DALL·E domain, the text-only RAC arm scores centered Vendi 77.67 and median nearest-neighbour distance 0.171; the vision-steered arm scores 91.97 and 0.192, an 18% gain in centered Vendi, and leads every column of the seven-arm comparison — distinct-2 0.699 against 0.660, 4-gram self-repetition 0.040 against 0.068, n -gram Vendi 170.9 against 167.0. The contact sheet of §5.2 shows the same thing without a number: rows 3–4, steered on both the text and the vision side, move medium, palette, register and composition where rows 1–2 return one visual mode.

6.2 Audio: cross-modal steering through an audio embedder

At matched n , matched prompt length and the same generator, the conditioned music arm reaches centered Vendi 43.03 against naive's 22.99 (1.87 $\times$), halves 4-gram self-repetition (0.140 against 0.291), raises distinct-2 (0.545 against 0.348) and holds roughly three times the room between nearest neighbours (0.079 against 0.027), on 100 prompts per arm with no exact duplicates in either — so the effect is semantic. On 100 rendered Lyria instrumentals generated by cross-modal steering with zero rejection, 100% of tracks verify as instrumental in CLAP space and mean-centered CLAP Vendi is 17.43 against 11.5 for naive prompts and 14.1 for axis-conditioned prompts under the same embedder.

The rendered arm was steered through CLAP and is measured in CLAP. §5.3's alignment table says the same loop through MuQ-MuLan would have a steering gradient nearly four times as strong on the naive arm (0.68 against 0.18), which is the experiment we would run next; the prescription — steer music with MuQ-MuLan rather than CLAP — is §5.3's, and the loop is the image one of §6.1 with the audio tower in place of CLIP.

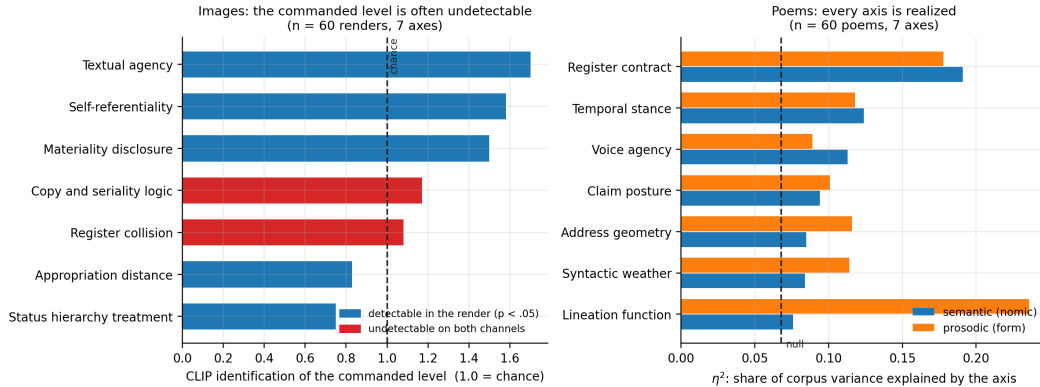
6.3 Best-of- K at the render: select on the channel the objective cannot see

The companion paper measures best-of- K on the written candidate (its §8.4): selecting on the gap alone raises the minimum nearest-neighbour distance by 20% in three of three seeds, at a cost of a third of a craft point. The same question at the *render* has a sharper answer, because the two channels have very different effective dimensions. Best-of- K on the written candidate happens before the handoff and §7 locates the loss after it, so we also ran $K = 3$ renders behind each of 60 fixed instructions and kept the render maximizing min distance to the accepted set — once in CLIP, once in optical statistics, over the same candidates. Selecting on optics raises pixel min NN from 1.039 to 1.292 (1.24 $\times$) and the fifth-percentile NN by 0.336 (paired subsample, 95% CI [+0.128, +0.488]), **and costs nothing in the channel the method optimizes**: CLIP min NN moves by -0.001 with the interval $[-0.029, +0.033]$ tight around zero. Selecting on CLIP raises CLIP min NN by 0.040 ([+0.025, +0.071]) and moves optics not at all detectably. Both gains lie on the $K^{1/m}$ curve — 1.24 $\times$ at $K = 3$ implies $m \approx 4.4$, and 1.04 $\times$ implies $m \approx 17$ — so render-side selection is oversampling rather than a new mechanism. What the dimensions add is an allocation rule: oversampling buys far more in a low-dimensional channel than in a high-dimensional one, and **if a pipeline is going to pay for extra renders, it should select them on the channel the objective cannot see**.

7. Conditioning across the seam

The companion paper audits whether a commanded axis level shows up in the artifact (its §8.1), by manipulation rather than correlation: hold a base spec fixed, set one axis to each of its levels in turn, generate, and read the displacement, centring within base so that only variation produced by *setting* the axis contributes, with a null that shuffles levels within each base. Two instruments read each artifact — *identifiability*, whether $\text{CLIP}(\ell)$ is the nearest of the axis's level descriptions to an image generated under level ℓ , against a chance rate of $1/|\text{levels}|$; and *separation*, the η^2 of artifact embeddings grouped by commanded level under a permutation null — and a third channel that lives outside the embedding the method optimizes: pixel statistics for images, prosody for poems. On poems every axis is realized on at least one channel. On images three of seven axes identify their level at or below chance, two are undetectable on both channels, and the axis the attribution scoring

ranked *first* is the least realized in the corpus. The image domain is where the proxy question becomes a conditioning question, because the image pipeline has a seam the poem pipeline does not.



*Figure 3. Axis realization measured on the artifact. Left: on 60 max-min renders, three of seven axes identify their commanded level at or below chance, and two are undetectable on both the CLIP and pixel channels. Right: on 60 poems, every axis is realized — on the semantic channel, the prosodic channel, or both. Dashed lines mark chance and the permutation null. **Images, by intervention: the loss is at the render, not in the writing.** The image pipeline has two stages — the model that proposed the axes writes an instruction, and gpt-image-1-mini renders it — so the same manipulation can be read on both sides of the handoff. Over all eleven axes at three bases and five levels, embedding the written instruction with CLIP, the render with CLIP, and the render's optics separately: nine of eleven axes are realized in the **written instruction** at $p < 0.05$ (mean $p = 0.148$), against four of eleven in the **render** (mean $p = 0.096$). Ten of the eleven attenuate, a mean loss of 35% of the text-side effect (Wilcoxon signed-rank $p = .005$; paired t , $p = .003$, both two-sided). **Conditioning reaches the instruction and loses a third of its grip at the render.** The image domain's weak realization is therefore not a failure of the axis calculus to produce usable conditioning — the conditioning is there, in the prompt, and measurable — but of that conditioning to survive a generator that never saw the axis set.*

Part of the apparent loss is the instrument rather than the renderer, and this is the §5.1 lesson arriving from the other direction. The four perceptual axes attenuate most in CLIP and are precisely the axes CLIP is least equipped to register: measured on pixels instead, *Lighting direction* reads 0.229 against 0.092 in CLIP, *Shot scale* 0.252 against 0.105, and *Motion and temporal blur* 0.203 against 0.069. An axis commanding optics moves the optics and barely moves a semantic embedding.

The seam explains the split. The calculus is identical across the two domains, the axes are comparably abstract, and the budgets match. What differs is where the conditioning has to travel. A poem is written by the same model that proposed the axes, so the conditioning never leaves the system that understands it. An image axis is written by that model into a prompt and handed to a different model, which never saw the axis set, shares no latent space with it, and is under no obligation to honour a distinction like *copies outrank an absent original*. §5.2 measured this seam from the outside as a proxy gap; the intervention measures it from the inside. **Conditioning strength degrades at the boundary where the generator changes hands**, so the modality-agnostic claim holds for the calculus — which needs only an embedding — and not for the conditioning, which needs a generator that reads the same language the axes are written in.

8. Scoring axes in the artifact's space

The companion paper scores a candidate axis by four factors — spread, transversality, independence and headroom (its §5.2) — all computed from *level vectors*: one embedding per level of each axis. In the text-only setting those are embeddings of the level descriptions — what a level *claims* it will do. Once the artifact can be embedded, the natural improvement is to replace them with the centroid of everything actually produced under each level: what the level *did*. That substitution is what makes the scoring empirical, and on its own it is wrong.

A centroid computed that way is a **marginal mean where the score needs a partial effect**. Specs are sampled independently, so every other axis varies inside each level group; with sixty items, seven axes and five levels each, a group holds about twelve items whose spread is mostly produced by the other six axes. The estimator cannot tell "this axis moves the artifact" from "this axis co-occurred with movement", and nothing in the loop ever intervenes on one axis while holding the rest fixed.

The consequence is not a modest loss of precision. It is a **systematic inversion**, and the mechanism is our own transversality term. An axis that genuinely moves the artifact fills the corpus with variance along its own direction; the occupied eigenspace absorbs that direction; and transversality — the fraction of an axis's variation lying *outside* the occupied span — collapses toward zero for precisely that axis. An axis the generator ignores contributes only isotropic noise, little of which the occupied basis captures, so its transversality stays high. Because promise is multiplicative, the effective axis sinks and the inert one rises. **The better an axis is, the harder the score punishes it**. The repair — partial effects from one additive model over all axes, a null-corrected realization factor ρ , and the interventional probe when a generation budget can be spent on measurement — is the companion paper's §5.3, and it is what makes artifact-space scoring usable: with it the strong axis ranks first on all 24 seeds of a ground-truth test where the attribution form ranked the inert axis first in 17 of 24. What belongs here is the consequence for a proxy. Level vectors read in the artifact's own embedding are only as good as that embedding's grip on the axis. §7 shows the four perceptual image axes attenuate most in CLIP and are precisely the axes CLIP is least equipped to register — *Lighting direction* reads 0.229 on pixels against 0.092 in CLIP — so an axis commanding optics can be scored as inert in the very space that was adopted to make the scoring honest. Where a designed feature channel exists for the artifact, the realization factor should be computed there as well, and the axis kept if either channel finds it.

9. Decode instead: the oracle is usually cheaper than the proxy

The argument for computing distances on text rather than on the artifact is always the same, and it is never stated as an argument because it is assumed: decoding is expensive, embedding is cheap, so one embeds. For a rendered image or a two-minute instrumental that is true. For the decoders in this section it is false, and by a wide margin.

Measured on one laptop, per item:

operation	ms per item	relative
decode: string $\rightarrow$ ipaddress arcs, plain execution	0.11	1×
decode: same, with sys.settrace instrumentation	0.98	9×
decode: Python program $\rightarrow$ 140-input battery, each in a fresh OS process	29.1	265×
embed: nomic-embed-text, local Ollama, batched	107.9	981×
embed: nomic-embed-text, local Ollama, one at a time	1,488.2	13,529×
embed: character TF-IDF, local, batched	18.3	166×

The code row is deliberately the least favourable decode measurement we can honestly make: every generated program is run in a *separate operating-system process* for isolation, so Python interpreter start-up dominates and the actual execution is a rounding error. Even so, decoding is **3.7× cheaper than a batched local embedding** and **51× cheaper than embedding items one at a time**, which is what a streaming pipeline does. Where the decoder does not need process isolation the margin is three orders of magnitude.

Two honest qualifications. Instrumentation, not the program, is the dominant decode cost — tracing costs 9× plain execution — so a decoder whose instrumentation is heavy needs its own measurement rather than an inherited ratio. And the embedding figures are for a *local* model with no network hop and no per-token charge, which is the best case for the text-space route; a hosted embedding endpoint is slower and has a price.

The conclusion is not rhetorical. **For a deterministic decoder that runs in milliseconds, the artifact-space oracle is not the expensive option a practitioner forgoes. It is the cheap option they did not think to run**, and it recovers between 1.02× and 1.64× the coverage of the best

text-space selector at matched budget (§4.6), reaching at $k = 30$ what text-space selection needs $k = 100$ to reach.

Where decoding genuinely is expensive — rendered images, audio, anything behind a paid endpoint — the fallback is not a text-space selector but a *partial decode budget*: decode a sample, and spend the rest of the budget where the sample says the behaviour is. That regime is where §6's cross-modal steering belongs, and it is the boundary between the two halves of this paper.

10. decodergap: the probe

The results above do not compose into a rule a practitioner can apply blind, because the one decision whose sign varies — far-field selection — varies *by decoder*, and nothing visible in the text says which case you are in. What they compose into is a measurement, and the measurement is cheap enough to run before any pipeline decision is committed.

decodergap takes three things: a corpus, an embedding function, and a decode function returning anything with a distance on it. It returns the near-field profile, the two deduplication verdicts, and the selector comparison against the artifact-space oracle.

```
from decodergap import probe

report = probe(
    texts      = corpus,                # what you generated
    embed      = my_embedder,           # what you were going to select in
    decode     = lambda s: run(s, battery), # what the item actually becomes
    distance   = jaccard,               # a distance on the decoded artifact
    coverage   = distinct_cells,        # what a subset is meant to buy
)

report.verdicts["deduplication_aggregate"] # SUPPORTED | UNSUPPORTED
report.verdicts["deduplication_per_item"]  # SUPPORTED | NOT SUPPORTED
report.verdicts["max_min_selection"]       # SAFE | UNSAFE, and vs the oracle
```

Three design decisions carry the tool, and each is a finding from above rather than a preference.

It never reports a correlation without its decoder-free control. The profile of §4.2 looks like a finding and is reproduced twice as strongly by two encoders that never saw a program run (§4.3). Any proxy-qualification number the tool emits is accompanied by the encoder-to-encoder baseline it has to beat, and by the permutation null that shows whether any association exists at all.

It issues two deduplication verdicts, not one. The aggregate verdict answers *can this radius rank duplicate pairs at all*; the per-item verdict answers *of the pairs a filter at this radius would reject, how many are behaviourally far apart*. On both decoders measured here the first passes and the second fails, at 45.4% and 41.6%. A tool that emitted a single "dedup: safe" would be licensing the decision its own data refuses.

It scores every selector in the artifact's space and against the oracle, so a text-space selector cannot win by agreeing with the representation it selected in, and beating random is not mistaken for doing well.

It also refuses to recommend a filter radius. We swept the threshold over the 5th to 40th percentiles of each corpus's own distance distribution at four budgets: the best cell on the `ipaddress` decoder is +20.7 arcs over random at $t = 10$ th percentile and $k = 10$, and by $k = 50$ every threshold loses. Any single-budget experiment would have produced a shippable-looking constant. The sweep is reported in full, including the cells that lose, and no constant is offered.

11. Practice

The prescriptions are short, and each is attached to the measurement that motivates it. The first four are the ones we would keep if we could keep four.

1. **Never report a proxy-validation correlation without a decoder-free control.** Correlate your embedding against a second, unrelated representation of the same corpus, and against a

permuted version of your artifact measurement. Whatever agreement survives above the first is what the artifact contributed; the second tells you whether any association exists. On our corpus the encoder-to-encoder baseline was **twice** the decoder signal (§4.3), which retires the finding we had.

2. **Qualify a proxy by running the decision, not by correlating the distances.** A correlation summarizes a geometry and inherits that geometry's confounds. A decision is scored in the artifact's space and either works or does not: what fraction of the pairs your filter drops are genuinely duplicates, how much artifact coverage your selector buys against the oracle.
3. **Split every deduplication claim into a corpus claim and an item claim.** They have different evidence and on both decoders here they get different verdicts: AUC 0.779 for ranking duplicate pairs, and 45.4% of rejected pairs behaviourally far apart at the tightest radius (§4.5). Contamination screening is defensible; a per-item gate is not.
4. **If your decoder is deterministic and fast, do not select in text space at all.** Decoding cost 29.1 ms per item against 107.9 ms to embed one, in the least favourable measurement we could construct (§9), and the artifact-space oracle reached at $k = 30$ what text-space selection needed $k = 100$ to reach (§4.6). The oracle is the cheap option, not the expensive one.
5. **Name the embedder, and publish the pairwise-similarity distribution it operates on.** Per-arm audio verdicts flip between embedders and each nominates a different most-redundant pair (§5.3); mean pairwise cosine was 0.883 in one text corpus and 0.444 in another under the same embedder. A diversity number without its embedder and its similarity distribution is not a number.
6. **Count exact duplicates before computing anything, and report the literal and the latent separately.** They move in opposite directions in the same corpus (§5.5), and a 74% duplicate rate makes every distance statistic a statistic about duplication.
7. **Measure at the level of the artifact you ship.** Text-embedding similarity explains 14.5% of the variance in whether rendered images look alike, pooled across seven arms, and 8.2% within-arm (§5.2). If the artifact is rendered, the objective belongs in the space the artifact occupies.
8. **Audit in at least one channel the objective never optimizes,** with a distance statistic rather than a spectral one (§5.1). It is the only measurement that cannot be a restatement of the selection rule, and it is where the strongest positive result and the largest undetected defect in this line of work were both found.
9. **When a render is in the loop, steer through it.** Render a bounded sample, embed it in the artifact's space, and bridge its crowded directions back into the text space the loop optimizes (§6.1); steer audio through an embedder whose text and audio towers agree (§6.2).
10. **Spend extra renders on the low-dimensional channel.** Best-of- K buys $K^{1/m}$; selecting renders on nine pixel statistics buys $1.24\times$ at $K = 3$ where selecting on CLIP buys $1.04\times$ (§6.3).
11. **Measure axis realization at the artifact, by intervention, wherever the generator changes hands** (§7). Where the seam exists, a third of the conditioning does not cross it.
12. **Balance by construction what the embedding cannot see.** Key position, difficulty, compositional template: none is encoded, none is optimized, and each decides usability. A uniform permutation of options needs no answer key and takes χ^2 from 90.4 to 2.6 (§5.7); a structural ban block halves palette twins (§5.6).
13. **Read the output.** A corpus can be diverse along every dimension you thought to measure and uniform along the one you did not.

12. Limitations

- **One text embedder.** All text geometry is nomic-embed-text with cosine distance. §5.4 shows the *ranking* of methods survives four representations; whether the *magnitudes* transfer is untested, and the companion paper's coverage and packing numbers inherit that.
- **A joint embedder varies both towers at once.** §5.3 changes one knob — the model supplying both the text tower and the audio tower — so it shows that the alignment moves and cannot say which side moves it. Crossing text-side and artifact-side encoders independently on fixed bytes is the decomposition this paper does not contain.

- **The renders carry no logged seed.** The image endpoint exposes no seed parameter, so no render in this paper can be regenerated, and render stochasticity cannot be separated from the quantity being measured. Every artifact-space number here — the proxy correlations of §5.2, the realization of §7, the render selection of §6.3 — is a statement about an unlogged seed distribution, and a repeat-draw floor at a logged seed is the missing measurement.
- **Several image results are single-seed.** The image arms are one seed each at $n = 60$; the vision-steered arm's margin in §6.1 is one run at $n = 200$; the audio steering arm is one run of 100 tracks.
- **Judge bias is unmeasured.** Vision and language-model judges name attractors, classify blueprint areas and read device forms; none is calibrated against human raters, and a judge that shares the generator's blind spots would miss exactly what the generator misses.
- **The structural signatures are deliberately dumb.** A 16×16 luminance map, a tiling score and a hue histogram find tilings and palette twins and nothing subtler; they are a floor on what the embedding misses, not a measure of it.
- **Steering closes part of the gap and not all of it.** Literal-space bans halve palette twins and improve the coverage objective by 21% and do not close the structural deficit; the vision-steered arm is the best arm in the image domain and still renders through a model that never saw the axes.

13. Conclusion

An embedding oracle is the only thing the loop can see, and it is always a proxy. In the two domains where the artifact is rendered by a model the embedder never meets, the proxy explains a seventh of the variance in what the reader receives, whether a prompt can be steered at all depends on which embedder is asked, and a third of the conditioning does not survive the handoff. In the domains where the artifact is text, the embedding is still blind to metre, to compositional template, and to where an answer key sits — properties that decide whether the corpus can be used.

None of this is an argument against the method; it is an argument about where the method's numbers are true. The extension that follows is to treat the proxy as a proxy: render a sample and steer through the artifact's own space, spend oversampling budget on the channel the objective cannot see, audit realization by intervention at the seam, and balance by construction what no embedding encodes. Under those rules the ranking the companion paper reports is robust — first under four representations, three of which the method never optimized, at a mean Kendall τ of +0.83 — and the gap between a diverse corpus and a usable one is at least visible, which is the precondition for closing it.